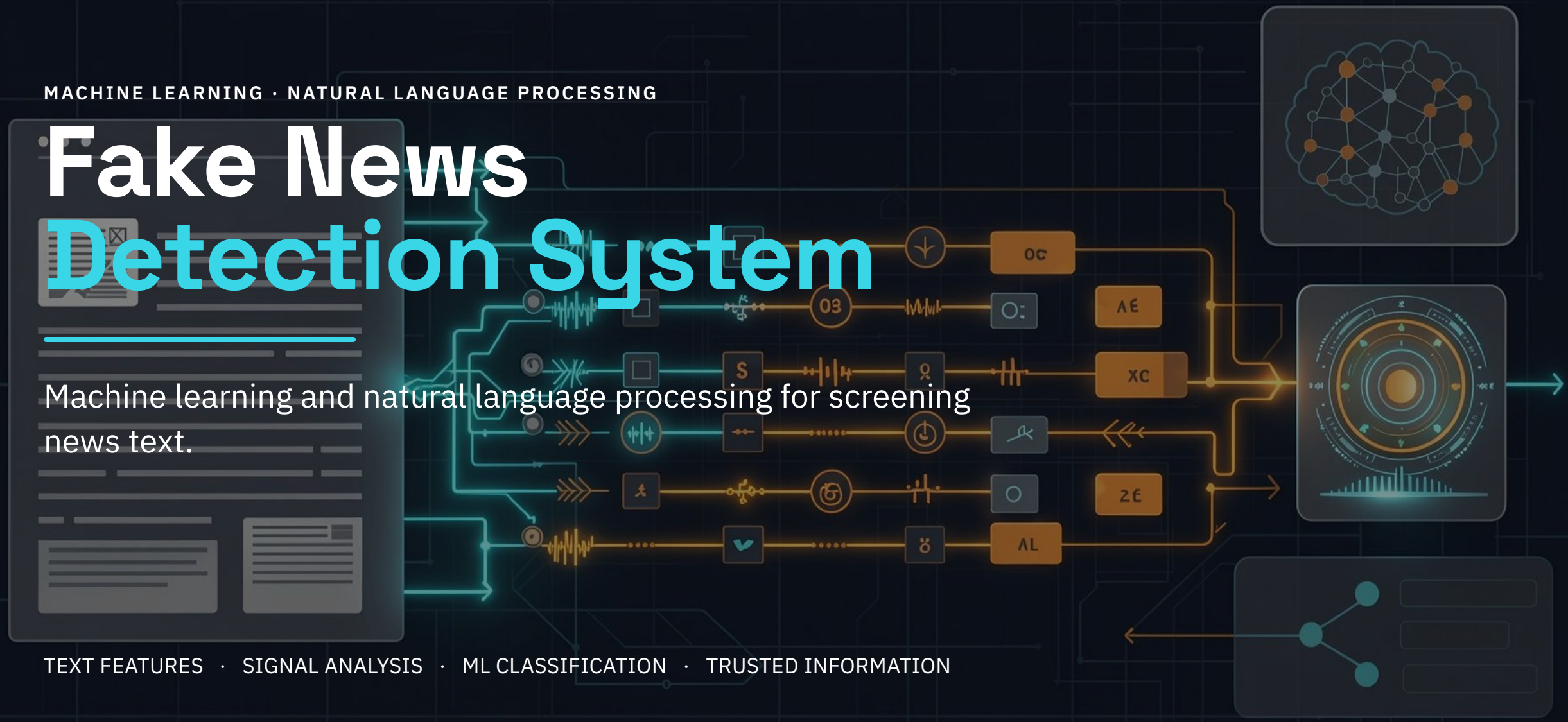
MACHINE LEARNING · NATURAL LANGUAGE PROCESSING

Fake News Detection System

Machine learning and natural language processing for screening news text.



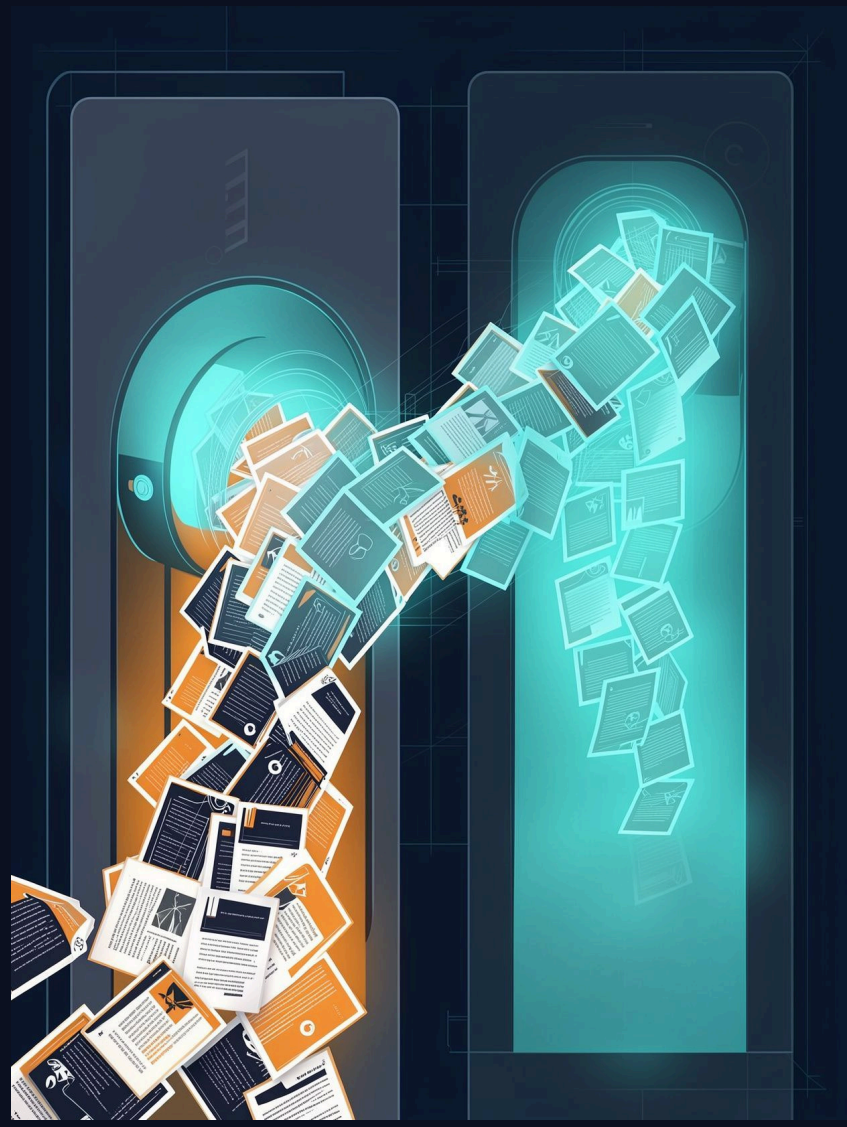
TEXT FEATURES · SIGNAL ANALYSIS · ML CLASSIFICATION · TRUSTED INFORMATION

Made with GAMMA

THE CHALLENGE

The challenge is fast, scalable news screening

- Digital platforms publish news at high volume.
- The system classifies an article as Fake News or Real News from its headline and body.
- It supports preliminary screening, not independent truth verification.



Preliminary screening at scale

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DATASET CONSTRUCTION

The project turns raw articles into a supervised learning dataset

SOURCE + INITIAL CONCATENATION

Fake.csv and True.csv

From the Kaggle dataset.

44,898 rows

CLEANED DATASET

39,103 labelled records

After removing empty and duplicate combined articles.

17,907 fake · 21,196 real examples

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SYSTEM WORKFLOW

The pipeline follows a clear path from text to prediction

Download, prepare, learn, evaluate, export, and infer.

01

Download & load

Load the dataset.

02

Combine text

Join title and body.

03

Clean

Normalize the text.

04

Split

Training and test sets.

05

Model & infer

Evaluate, export, and infer.

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TEXT PREPARATION

Text preparation removes noise without changing the message

CLEANING

Text preparation

- Lowercase text and remove HTML tags.
- Replace URLs and email addresses with standard tokens.
- Collapse repeated whitespace.

DATA SPLIT

Stratified 80/20 split

TRAINING

31,282

records

TEST

7,821

records

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FEATURE EXTRACTION

TF-IDF captures words and short phrases that matter

The representation keeps useful local signal while protecting the held-out test set.



Unigrams

Individual terms become features.



Bigrams

Adjacent phrases preserve short context.



Filtered vocabulary

English stop words are excluded.

VOCABULARY CAP

100,000 features

LEAKAGE CONTROL

Training text only

The vectorizer is fitted on training text only.

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MODEL COMPARISON

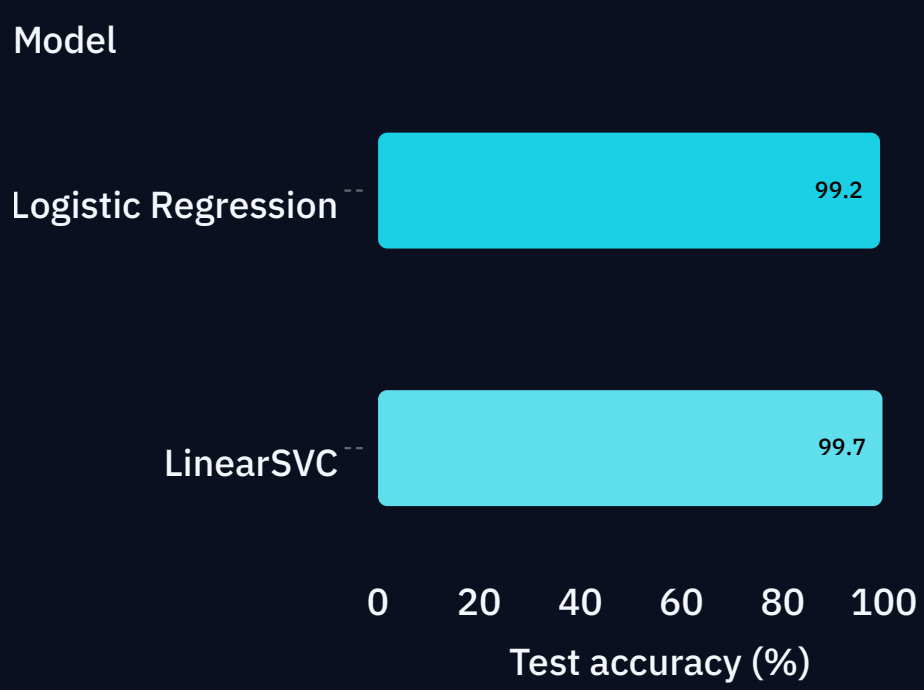
LinearSVC gives the strongest recorded baseline

Both classifiers use the same prepared text representation.

SELECTED FOR EXPORT

LinearSVC

99.69% test accuracy



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RECORDED TEST RESULT

The recorded test result is 99.69% on this holdout

TEST ACCURACY

99.69%

on this holdout

TEST SET

7,821 examples

FAKE CLASS SUPPORT · REAL CLASS SUPPORT

3,582 · 4,239

ROUNDED METRICS

1.00

Precision, recall, and F1 values display as 1.00 for both classes.

Scope: the score describes performance on this dataset split, not universal truth detection.

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REAL-TIME INFERENCE

One saved pipeline connects training to real-time inference

01

Input

Headline and article body.

02

Same cleaning

Apply the training-time text preparation.

03

Saved pipeline

TF-IDF plus the selected classifier in one Joblib pipeline.

04

Output

FAKE NEWS or REAL NEWS

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REPRODUCIBLE WORKFLOW

Google Colab makes the workflow reproducible

01 Install and download

Install packages and download the Kaggle dataset.

02 Prepare the data

Load, clean, split, and vectorize the data.

03 Compare classifiers

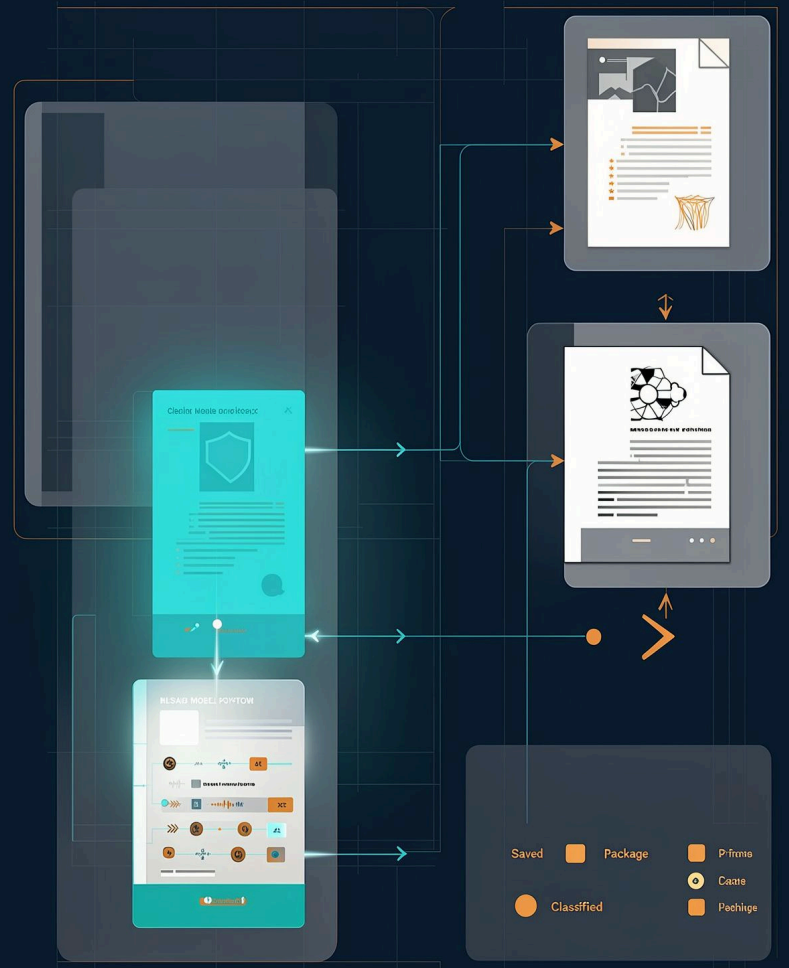
Train both classifiers and compare metrics.

04 Save and test

Save the winning pipeline and test it on real examples.

05 Download for reuse

Download the model for later reuse.



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